

Nithin Venkat Sharma P M

Coimbatore, Tamil Nadu

✉ nithin.p.m2006@gmail.com

☎ +91 73730 93900

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🔗 nithino306

SUMMARY

Computer Science undergraduate with hands-on experience spanning full-stack software development, data structures, and AI system architecture. Skilled in integrating modern web frameworks with tool-calling protocols and AI pipelines to automate complex workflows. Highly motivated to collaborate in team environments, writing clean code to build scalable, intelligent, and user-focused software solutions.

EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore

Aug 2024 – May 2028

B.Tech in Computer Science

- CGPA: 8.85 / 10

EXPERIENCE

Web Developer - Tensor Club, Amrita Vishwa Vidyapeetham

Jul 2025 – Apr 2026

- Contributed to frontend development of the club website, improving UI responsiveness and usability
- Collaborated with team members to implement layout and styling enhancements based on feedback

Automation Intern - The Pedagogy Community

Apr 2026 – Present

- Built AI pipeline to convert webinar videos into structured courses using Whisper, GPT-4, FastAPI, and Next.js
- Collaborated with mentors to optimize prompts, validate output quality, and ship MVP to real users

PROJECTS

Dayta | React (Vite, TypeScript), FastAPI, Pinecone, Google Gemini API

[Live](#)

- Built a full-stack Progressive Web App integrating a RAG pipeline to automate and accurately resolve academic schedule queries.
- Architected the complete AI workflow—from document ingestion to final Answer Generation using Pinecone vector search for low-latency retrieval.
- Implemented strict conversational guardrails and session memory, confining responses entirely to official university data to prevent LLM hallucinations.

Archivist | Python, LangGraph, GitHub Actions, LLMs, MCP GitHub

[GitHub](#)

- Developed an autonomous CI/CD GitHub Action using LangGraph to automatically evaluate Pull Request diffs against live Architecture Decision Records (ADRs).
- Utilized the Model Context Protocol (MCP) to dynamically ingest architectural guidelines and detect framework-specific code violations pre-merge.
- Enforced enterprise-level code governance by blocking unauthorized merges via the GitHub API, reducing technical debt while supporting a secure developer override workflow.

GoCDC (Event-Driven CDC Pipeline) | Go, PostgreSQL, Apache Kafka, Redis, MongoDB, Docker

[GitHub](#)

- Engineered a high-performance Change Data Capture (CDC) engine in Go, directly intercepting PostgreSQL Write-Ahead Logs (WAL) via logical decoding for sub-millisecond data replication.
- Architected a fault-tolerant, decoupled microservice ecosystem utilizing Apache Kafka to broadcast standardized JSON event payloads, guaranteeing zero data loss during simulated system failures.
- Implemented parallel consumer routines to achieve polyglot persistence, synchronously fanning out relational DML operations into a high-speed Redis cache and a MongoDB document store.

TECHNICAL SKILLS

- **Languages:** C, C++, Java, JavaScript, Python, SQL, Go (Golang)
- **Frameworks & Libraries:** React.js, FastAPI, Node.js, Express.js, LangChain, LangGraph
- **Databases:** PostgreSQL, MongoDB, Supabase
- **Tools & Technologies:** Git, GitHub, Docker, Postman, Linux
- **Core Concepts:** Data Structures, Problem Solving, Software Development Lifecycle (SDLC), Team Collaboration

ACHIEVEMENTS

- Secured 2nd place at Hack101 for SkillUp, a team-based resume analysis and learning recommendation platform.
- Secured 3rd place in the Build2Break hackathon at Anokha 2026 for developing Vita-Care